

Class 27 Math Induction

Goals

1. Learn the difference between weak and strong induction.
2. Practice several examples.

Example 1: Prove that $7^n - 1$ is a multiple of 6 for all $n \geq 1$

$P(n)$ is the statement " $7^n - 1 = 6X$ for some integer X ."

Base case: $P(1)$ is true since $7^1 - 1 = 6 = 6 \cdot 1$

Induction Hypothesis: Assume $P(k)$ is true.

show $P(k+1)$ is true

Let $7^k - 1 = 6X$ for some $X \in \mathbb{Z}$.

$$\begin{aligned} \text{So } 7^{k+1} - 1 &= 7 \cdot 7^k - 1 = 6 \cdot 7^k + \underbrace{7^k - 1} \\ &= 6 \cdot 7^k + 6X = 6(7^k + X) \text{ is a multiple} \\ &\text{of 6 since } 7^k + X \text{ is an integer.} \end{aligned}$$

Thus $P(k+1)$ is true

Conclusion: Since the statement is true for $n=1$ and is true for $n=k+1$ whenever true for $n=k \geq 1$ we know the statement is true for all integers $n \geq 1$

Strong Induction: If $[P(a)$ is true for some starting integer a and if all the statements from $P(a)$ through $P(k)$ being true implies that $P(k+1)$ is true], then $P(n)$ is true for all $n \geq a$.

Example 2: Use strong induction to prove every integer greater than 1 is a prime or a product of primes. (existence part of the Fundamental Theorem of Arithmetic).

$P(n)$ is the statement " n is prime or a product of primes"

Base case $P(2)$ is true since 2 is prime.

Induction Hypothesis: Assume $P(n)$ is true for $2 \leq n \leq k$

So every integer from 2 to k is either prime or a product of primes

Show $P(k+1)$ is true. That is, show $k+1$ is either prime or a product of primes.

If $k+1$ is prime, then $P(k+1)$ is true.

If $k+1$ is not prime, write $k+1 = a \cdot b$

where a and b are integers with $2 \leq a, b \leq k$

So by the I.H. a and b are primes or products of primes.

This means $k+1 = a \cdot b$ is a product of primes so $P(k+1)$ is true.

Conclusion: Since the statement is true for $n = 2$ and is true for $n = k+1$ whenever true for $n = 2$ through k , we know the statement is true for all integers $n \geq 2$

Example 3: The Fibonacci sequence is defined recursively by $f_1 = 1$, $f_2 = 1$, and $f_{n+2} = f_n + f_{n+1}$ for $n \geq 1$. Write out the first 8 terms of the sequence.

Use strong induction to prove $f_n < 2^n$ for all $n \geq 1$.

Base case $f_1 = 1 < 2^1 \checkmark$ and $f_2 = 1 < 2^2$. (We need two terms since the recursion involves 2 terms.)
So $P(1)$ and $P(2)$ are both true.

Let k be any integer ≥ 1 , and assume $P(1), P(2), \dots, P(k)$ are all true. Show $P(k+1)$ is true

So we know $f_1 < 2^1, f_2 < 2^2, \dots, f_{k-1} < 2^{k-1}, f_k < 2^k$

$$f_{k+1} = f_{k-1} + f_k < 2^{k-1} + 2^k < 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$$

Conclusion: Since the statement is true for $n = 1$ and $n = 2$ and is true for $n = k+1$ whenever true for $n = 1$ through k , we know by strong induction the statement is true for all integers $n \geq 1$.

Example 4: Prove for all $n \geq 1$ $\prod_{i=1}^n \frac{1}{(2i+1)(2i+2)} = \frac{2}{(2n+2)!}$

Base case $P(1)$ is true since $\prod_{i=1}^1 \frac{1}{(2i+1)(2i+2)} = \frac{1}{3 \cdot 4} = \frac{1}{12}$ and

$$\frac{2}{(2 \cdot 1 + 2)!} = \frac{2}{4!} = \frac{2}{24} = \frac{1}{12}$$

Induction Hypothesis Let $\prod_{i=1}^k \frac{1}{(2i+1)(2i+2)} = \frac{2}{(2k+2)!}$ for $k \geq 1$

$$\begin{aligned} \text{This means } \prod_{i=1}^{k+1} \frac{1}{(2i+1)(2i+2)} &= \prod_{i=1}^k \frac{1}{(2i+1)(2i+2)} \cdot \frac{1}{(2(k+1)+1)(2(k+1)+2)} = \frac{2}{(2k+2)!} \cdot \frac{1}{(2k+3)(2k+4)} \\ &= \frac{2}{(2k+4)!} = \frac{2}{(2(k+1)+2)!} \quad \text{so } P(k+1) \text{ is true.} \end{aligned}$$

Conclusion: Since the statement is true for $n = 1$ and is true for $n = k+1$ whenever true for $n = k$, we know by math induction the statement is true for all integers $n \geq 1$.

